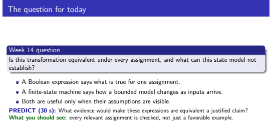
Boolean Algebra, Circuits, SAT, and State Models

Exhaust small cases; name what the model leaves out

COMSC 2043 ▪ Discrete Structures and Critical Thinking

Week 14, Tuesday, November 17, 2026

Tuesday is 75 minutes. Keep the rhythm: predict, run or fill the evidence, explain what survived, and name the boundary.



The question for today

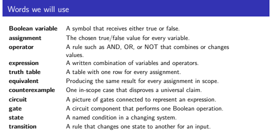
Week 14 question

Is this transformation equivalent under every assignment, and what can this state model not establish?

- A Boolean expression says what is true for one assignment.
- A finite-state machine says how a bounded model changes as inputs arrive.
- Both are useful only when their assumptions are visible.

PREDICT (30 s): What evidence would make these expressions are equivalent a justified claim?
What you should see: every relevant assignment is checked, not just a favorable example.

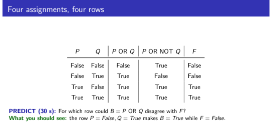
Connect the two halves: exhaustive rows for Boolean claims, explicit traces and transitions for state claims.



Words we will use

Boolean variable	A symbol that receives either true or false.
assignment	The chosen true/false value for every variable.
operator	A rule such as AND, OR, or NOT that combines or changes values.
expression	A written combination of variables and operators.
truth table	A table with one row for every assignment.
equivalent	Producing the same result for every assignment in scope.
counterexample	One in-scope case that disproves a universal claim.
circuit	A picture of gates connected to represent an expression.
gate	A circuit component that performs one Boolean operation.
state	A named condition in a changing system.
transition	A rule that changes one state to another for an input.

Do not read every line aloud. Point out that the terms are defined before we rely on them.



Four assignments, four rows

P	Q	P OR Q	P OR NOT Q	F
False	False	False	True	False
False	True	True	False	False
True	False	True	True	True
True	True	True	True	True

PREDICT (30 s): For which row could $B = P$ OR Q disagree with F?
What you should see: the row $P =$ *False*, $Q =$ *True* makes $B =$ *True* while $F =$ *False*.

Fill the row values with the class if possible. The point is not speed; it is covering the entire finite space.

Prompt 01: make the claim checkable

PROMPT FOR CLAUDE prompts/01_session_rules.md

Act as a careful checking partner for a two-variable Boolean example. Before writing code, state what an expression, a Boolean variable, an assignment, and equivalence mean in plain language. Use only the operators AND, OR, and NOT. Keep the made-up example self-contained. Then propose the smallest Python standard-library evaluator you would need for a truth table. Do not claim two expressions are equivalent until every assignment has been checked. Show one predicted row before showing the rest.

PREDICT (30 s): What must be defined before an evaluator can be trusted?
What you should see: terms, operators, assignment order, and the meaning of equivalence are visible.

Have students identify a missing definition before running anything.

Exit sentence

Complete this sentence

For this declared model, the evidence establishes ____________________; it does not establish ____________________ because _______________.

- A truth table establishes behavior for the assignments it exhausts.
- A trace establishes the path for the inputs and transitions it names.
- A model limit protects the claim from becoming larger than its evidence.

Point students toward the canonical Checkpoint 3 source for Thursday; this deck does not reproduce that source.

Index 1 of 3: principle, slide, prompt

Slide	Title	Principle	Prompt
2	The question for today	MODEL SCOPE	
3	Our checking rhythm	PREDICT–RUN–EXPLAIN	
5	The tempting simplification	CHECK THE CLAIM	
6	Four assignments, four rows	EXHAUST SMALL CASES	
7	One counterexample is enough	COUNTEREXAMPLE	
8	Prompt 01: make the claim checkable	DEFINE BEFORE COMPUTING	01a
9	Prompt 02: make every row visible	EXHAUST SMALL CASES	02a
10	A correct simplification from the activity	COMPARE CLAIMS	
11	From expression to circuit	REPRESENTATION	

Use this index to find a slide by number, principle, and prompt file.

Index 2 of 3: principle, slide, prompt

Slide	Title	Principle	Prompt
12	Equivalence is a universal statement	MATCH THE SCOPE	
13	Satisfiable is not always true	SEPARATE CLAIMS	
14	Three small classifications	CHECK THE WORDING	
15	Prompt 03a: return counterexamples	TEST THE UNIVERSAL CLAIM	03a
16	Prompt 03b: witness versus tautology	DO NOT OVERCLAIM	03b
17	Truth tables and circuits agree when the representation agrees	CHECK THE REPRESENTATION	
18	Now change the question: how does it move?	FROM VALUES TO STATES	
19	Synthetic badge model	MAKE TRANSITIONS EXPLICIT	
20	Trace it one input at a time	CHECK THE TRACE	

Use this index to find a slide by number, principle, and prompt file.

Index 3 of 3: principle, slide, prompt

Slide	Title	Principle	Prompt
21	Prompt 04: run the complete model	REPRODUCIBLE TRACE	04a
22	A model can be incomplete	FIND WHAT WAS FORGOTTEN	
23	Prompt 05: name the missing transition	VALIDATE THE MODEL	05a
24	What the badge model cannot establish	MODEL LIMIT	
25	The three AI failure modes	FAILURE MODES	
26	Tuesday individual activity	MAKE YOUR OWN EVIDENCE	
27	Preview: defend a bounded model	EVIDENCE BEFORE CONFIDENCE	
28	Exit sentence	LIMIT THE CLAIM	
29	One last check before trust	CHECK THE WORK	

Use this index to find a slide by number, principle, and prompt file.